

Planetary Gearbox - 3-Stage Teardown and Repair Analysis

A. Job information

|                                       |                          |               |               |
|---------------------------------------|--------------------------|---------------|---------------|
| Customer name                         | Work order number        |               |               |
|                                       |                          |               |               |
| Date received                         | Manufacturer             |               |               |
|                                       |                          |               |               |
| Model                                 | Serial number            |               |               |
|                                       |                          |               |               |
| Overall reduction                     | Stage 1 ratio            | Stage 2 ratio | Stage 3 ratio |
| : 1                                   | : 1                      | : 1           | : 1           |
| Oil capacity                          | Oil grade                |               |               |
|                                       |                          |               |               |
| Application / driven equipment        | Service hours (if known) |               |               |
|                                       |                          |               |               |
| Reported failure / customer complaint |                          |               |               |
|                                       |                          |               |               |

B. As-received condition

|                          |                                        |
|--------------------------|----------------------------------------|
| <input type="checkbox"/> | Photos taken as received               |
|                          | Notes                                  |
|                          |                                        |
| <input type="checkbox"/> | External damage, cracks, missing parts |
|                          | Notes                                  |
|                          |                                        |
| <input type="checkbox"/> | Nameplate present and legible          |
|                          | Notes                                  |
|                          |                                        |

# Ram Gear Gearbox Evaluation

Shafts turn freely by hand: ☐ Yes ☐ No

Notes

☐ Noise or roughness when turned

Notes

☐ Oil leaks at seals / split line / plugs

Notes

☐ Breather condition

Notes

## C. Oil condition

Oil drained - volume recovered

Oil sample ID (lab analysis)

☐ Oil sample taken for lab analysis

Appearance: ☐ Clean ☐ Dark ☐ Milky (water) ☐ Sludge ☐ Metallic particles

Magnetic drain plug debris description

## D. As-found measurements before teardown

Endplay - Input shaft

Endplay - Output shaft

Output rotational backlash (input locked)

Ext. runout - Input shaft

Ext. runout - Output shaft

# Ram Gear Gearbox Evaluation

## E. Teardown checklist

- ☐ Match-mark housing, ring gears, carriers and shafts before disassembly
- ☐ Parts tagged and kept in order, by planetary stage
- ☐ Seals removed and inspected

## As-found shim record

| Shaft  | Drive end (DE) as-found shim (in/mm) | Non-drive end (NDE) as-found shim (in/mm) | Shim pack                        |                                |
|--------|--------------------------------------|-------------------------------------------|----------------------------------|--------------------------------|
| Input  | <input type="text"/>                 | <input type="text"/>                      | <input type="checkbox"/> Replace | <input type="checkbox"/> Reuse |
| Output | <input type="text"/>                 | <input type="text"/>                      | <input type="checkbox"/> Replace | <input type="checkbox"/> Reuse |

## P1. Planetary stage 1

|                                              |                          |                                                      |                      |
|----------------------------------------------|--------------------------|------------------------------------------------------|----------------------|
| Sun gear teeth (Z sun)                       | Ring gear teeth (Z ring) | Planet gear teeth                                    | Number of planets    |
| <input type="text"/>                         | <input type="text"/>     | <input type="text"/>                                 | <input type="text"/> |
| Stage ratio, calculated (1 + Z ring / Z sun) |                          | Stage ratio, measured (input turns per carrier turn) |                      |
| <input type="text"/> : 1                     |                          | <input type="text"/> : 1                             |                      |

## Stage 1 measurements

| Measurement                 | Measured             | Spec                 | OK                       |
|-----------------------------|----------------------|----------------------|--------------------------|
| Sun-planet backlash         | <input type="text"/> | <input type="text"/> | <input type="checkbox"/> |
| Planet-ring backlash        | <input type="text"/> | <input type="text"/> | <input type="checkbox"/> |
| Sun gear float / axial play | <input type="text"/> | <input type="text"/> | <input type="checkbox"/> |
| Carrier endplay             | <input type="text"/> | <input type="text"/> | <input type="checkbox"/> |
| Thrust washer thickness     | <input type="text"/> | <input type="text"/> | <input type="checkbox"/> |

## Sun gear

Sun gear condition: ☐ No defects ☐ Tooth wear ☐ Pitting / spalling  
☐ Spline wear / fretting ☐ Cracks / breakage

# Ram Gear Gearbox Evaluation

## Stage 1 sun gear

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

## Planet gears

### Stage 1 planets (one row per planet)

| Planet   | Gear teeth condition | Pin / shaft wear & fit | Bearing / needle rollers | Thrust washers       | Planet                         |                                 |                                  |
|----------|----------------------|------------------------|--------------------------|----------------------|--------------------------------|---------------------------------|----------------------------------|
| Planet 1 | <input type="text"/> | <input type="text"/>   | <input type="text"/>     | <input type="text"/> | <input type="checkbox"/> Reuse | <input type="checkbox"/> Repair | <input type="checkbox"/> Replace |
| Planet 2 | <input type="text"/> | <input type="text"/>   | <input type="text"/>     | <input type="text"/> | <input type="checkbox"/> Reuse | <input type="checkbox"/> Repair | <input type="checkbox"/> Replace |
| Planet 3 | <input type="text"/> | <input type="text"/>   | <input type="text"/>     | <input type="text"/> | <input type="checkbox"/> Reuse | <input type="checkbox"/> Repair | <input type="checkbox"/> Replace |
| Planet 4 | <input type="text"/> | <input type="text"/>   | <input type="text"/>     | <input type="text"/> | <input type="checkbox"/> Reuse | <input type="checkbox"/> Repair | <input type="checkbox"/> Replace |
| Planet 5 | <input type="text"/> | <input type="text"/>   | <input type="text"/>     | <input type="text"/> | <input type="checkbox"/> Reuse | <input type="checkbox"/> Repair | <input type="checkbox"/> Replace |
| Planet 6 | <input type="text"/> | <input type="text"/>   | <input type="text"/>     | <input type="text"/> | <input type="checkbox"/> Reuse | <input type="checkbox"/> Repair | <input type="checkbox"/> Replace |

## Stage 1 planet pins / shafts

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

## Stage 1 planet bearings / needle rollers

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

## Stage 1 thrust washers

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

## Ring gear

Ring gear condition: ☐ No defects ☐ Internal teeth wear ☐ Pitting / spalling ☐ Cracks  
☐ Mounting / housing fit worn ☐ Dowels / bolts loose or damaged

# Ram Gear Gearbox Evaluation

## Stage 1 ring gear

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

## Planet carrier

Carrier condition: ☐ No defects ☐ Cracks ☐ Pin bores worn / oval  
☐ Carrier bearing damage ☐ Spline / output coupling wear

## Stage 1 planet carrier

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

## Stage 1 carrier bearing

☐ Reuse ☐ Repair ☐ Replace

Cone P/N:

Cup P/N:

Findings:

## Stage 1 coupling / spline, stage 1 to stage 2

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

## P2. Planetary stage 2

Sun gear teeth (Z sun)

Ring gear teeth (Z ring)

Planet gear teeth

Number of planets

Stage ratio, calculated ( $1 + Z \text{ ring} / Z \text{ sun}$ )

 : 1

Stage ratio, measured (input turns per carrier turn)

 : 1

## Stage 2 measurements

| Measurement                 | Measured             | Spec                 | OK                       |
|-----------------------------|----------------------|----------------------|--------------------------|
| Sun-planet backlash         | <input type="text"/> | <input type="text"/> | <input type="checkbox"/> |
| Planet-ring backlash        | <input type="text"/> | <input type="text"/> | <input type="checkbox"/> |
| Sun gear float / axial play | <input type="text"/> | <input type="text"/> | <input type="checkbox"/> |
| Carrier endplay             | <input type="text"/> | <input type="text"/> | <input type="checkbox"/> |
| Thrust washer thickness     | <input type="text"/> | <input type="text"/> | <input type="checkbox"/> |

# Ram Gear Gearbox Evaluation

## Sun gear

Sun gear condition: ☐ No defects ☐ Tooth wear ☐ Pitting / spalling  
☐ Spline wear / fretting ☐ Cracks / breakage

### Stage 2 sun gear

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

## Planet gears

### Stage 2 planets (one row per planet)

| Planet   | Gear teeth condition | Pin / shaft wear & fit | Bearing / needle rollers | Thrust washers       | Planet                         |                                 |                                  |
|----------|----------------------|------------------------|--------------------------|----------------------|--------------------------------|---------------------------------|----------------------------------|
| Planet 1 | <input type="text"/> | <input type="text"/>   | <input type="text"/>     | <input type="text"/> | <input type="checkbox"/> Reuse | <input type="checkbox"/> Repair | <input type="checkbox"/> Replace |
| Planet 2 | <input type="text"/> | <input type="text"/>   | <input type="text"/>     | <input type="text"/> | <input type="checkbox"/> Reuse | <input type="checkbox"/> Repair | <input type="checkbox"/> Replace |
| Planet 3 | <input type="text"/> | <input type="text"/>   | <input type="text"/>     | <input type="text"/> | <input type="checkbox"/> Reuse | <input type="checkbox"/> Repair | <input type="checkbox"/> Replace |
| Planet 4 | <input type="text"/> | <input type="text"/>   | <input type="text"/>     | <input type="text"/> | <input type="checkbox"/> Reuse | <input type="checkbox"/> Repair | <input type="checkbox"/> Replace |
| Planet 5 | <input type="text"/> | <input type="text"/>   | <input type="text"/>     | <input type="text"/> | <input type="checkbox"/> Reuse | <input type="checkbox"/> Repair | <input type="checkbox"/> Replace |
| Planet 6 | <input type="text"/> | <input type="text"/>   | <input type="text"/>     | <input type="text"/> | <input type="checkbox"/> Reuse | <input type="checkbox"/> Repair | <input type="checkbox"/> Replace |

### Stage 2 planet pins / shafts

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

### Stage 2 planet bearings / needle rollers

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

### Stage 2 thrust washers

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

## Ring gear

Ring gear condition: ☐ No defects ☐ Internal teeth wear ☐ Pitting / spalling ☐ Cracks

# Ram Gear Gearbox Evaluation

☐ Mounting / housing fit worn ☐ Dowels / bolts loose or damaged

## Stage 2 ring gear

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

## Planet carrier

Carrier condition: ☐ No defects ☐ Cracks ☐ Pin bores worn / oval

☐ Carrier bearing damage ☐ Spline / output coupling wear

## Stage 2 planet carrier

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

## Stage 2 carrier bearing

☐ Reuse ☐ Repair ☐ Replace

Cone P/N:

Cup P/N:

Findings:

## Stage 2 coupling / spline, stage 2 to stage 3

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

## P3. Planetary stage 3

Sun gear teeth (Z sun)

Ring gear teeth (Z ring)

Planet gear teeth

Number of planets

Stage ratio, calculated (1 + Z ring / Z sun)

 : 1

Stage ratio, measured (input turns per carrier turn)

 : 1

## Stage 3 measurements

| Measurement                 | Measured             | Spec                 | OK                       |
|-----------------------------|----------------------|----------------------|--------------------------|
| Sun-planet backlash         | <input type="text"/> | <input type="text"/> | <input type="checkbox"/> |
| Planet-ring backlash        | <input type="text"/> | <input type="text"/> | <input type="checkbox"/> |
| Sun gear float / axial play | <input type="text"/> | <input type="text"/> | <input type="checkbox"/> |
| Carrier endplay             | <input type="text"/> | <input type="text"/> | <input type="checkbox"/> |

Obtain all dimensional, endplay, backlash, and clearance specs from OEM drawings and bearing manufacturer data.

Form: gearbox-teardown-analysis-planetary3

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# Ram Gear Gearbox Evaluation

| Measurement             | Measured             | Spec                 | OK                       |
|-------------------------|----------------------|----------------------|--------------------------|
| Thrust washer thickness | <input type="text"/> | <input type="text"/> | <input type="checkbox"/> |

## Sun gear

Sun gear condition: ☐ No defects ☐ Tooth wear ☐ Pitting / spalling  
☐ Spline wear / fretting ☐ Cracks / breakage

|                                |                                |                                 |                                  |
|--------------------------------|--------------------------------|---------------------------------|----------------------------------|
| Stage 3 sun gear               | <input type="checkbox"/> Reuse | <input type="checkbox"/> Repair | <input type="checkbox"/> Replace |
| Part no.: <input type="text"/> | Findings: <input type="text"/> |                                 |                                  |

## Planet gears

### Stage 3 planets (one row per planet)

| Planet   | Gear teeth condition | Pin / shaft wear & fit | Bearing / needle rollers | Thrust washers       | Planet                         |                                 |                                  |
|----------|----------------------|------------------------|--------------------------|----------------------|--------------------------------|---------------------------------|----------------------------------|
| Planet 1 | <input type="text"/> | <input type="text"/>   | <input type="text"/>     | <input type="text"/> | <input type="checkbox"/> Reuse | <input type="checkbox"/> Repair | <input type="checkbox"/> Replace |
| Planet 2 | <input type="text"/> | <input type="text"/>   | <input type="text"/>     | <input type="text"/> | <input type="checkbox"/> Reuse | <input type="checkbox"/> Repair | <input type="checkbox"/> Replace |
| Planet 3 | <input type="text"/> | <input type="text"/>   | <input type="text"/>     | <input type="text"/> | <input type="checkbox"/> Reuse | <input type="checkbox"/> Repair | <input type="checkbox"/> Replace |
| Planet 4 | <input type="text"/> | <input type="text"/>   | <input type="text"/>     | <input type="text"/> | <input type="checkbox"/> Reuse | <input type="checkbox"/> Repair | <input type="checkbox"/> Replace |
| Planet 5 | <input type="text"/> | <input type="text"/>   | <input type="text"/>     | <input type="text"/> | <input type="checkbox"/> Reuse | <input type="checkbox"/> Repair | <input type="checkbox"/> Replace |
| Planet 6 | <input type="text"/> | <input type="text"/>   | <input type="text"/>     | <input type="text"/> | <input type="checkbox"/> Reuse | <input type="checkbox"/> Repair | <input type="checkbox"/> Replace |

|                                |                                |                                 |                                  |
|--------------------------------|--------------------------------|---------------------------------|----------------------------------|
| Stage 3 planet pins / shafts   | <input type="checkbox"/> Reuse | <input type="checkbox"/> Repair | <input type="checkbox"/> Replace |
| Part no.: <input type="text"/> | Findings: <input type="text"/> |                                 |                                  |

|                                          |                                |                                 |                                  |
|------------------------------------------|--------------------------------|---------------------------------|----------------------------------|
| Stage 3 planet bearings / needle rollers | <input type="checkbox"/> Reuse | <input type="checkbox"/> Repair | <input type="checkbox"/> Replace |
| Part no.: <input type="text"/>           | Findings: <input type="text"/> |                                 |                                  |

|                                |                                |                                 |                                  |
|--------------------------------|--------------------------------|---------------------------------|----------------------------------|
| Stage 3 thrust washers         | <input type="checkbox"/> Reuse | <input type="checkbox"/> Repair | <input type="checkbox"/> Replace |
| Part no.: <input type="text"/> | Findings: <input type="text"/> |                                 |                                  |

# Ram Gear Gearbox Evaluation

## Ring gear

Ring gear condition: ☐ No defects ☐ Internal teeth wear ☐ Pitting / spalling ☐ Cracks  
☐ Mounting / housing fit worn ☐ Dowels / bolts loose or damaged

### Stage 3 ring gear

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

## Planet carrier

Carrier condition: ☐ No defects ☐ Cracks ☐ Pin bores worn / oval  
☐ Carrier bearing damage ☐ Spline / output coupling wear

### Stage 3 planet carrier

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

### Stage 3 carrier bearing

☐ Reuse ☐ Repair ☐ Replace

Cone P/N:

Cup P/N:

Findings:

### Stage 3 output spline / coupling (stage 3 carrier to output)

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

## F. Input, output, housing and lubrication

### Input shaft

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

### Input seal

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

### Input shaft bearings

☐ Reuse ☐ Repair ☐ Replace

Cone P/N:

Cup P/N:

Findings:

# Ram Gear Gearbox Evaluation

## Output shaft

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

## Output shaft bearings

☐ Reuse ☐ Repair ☐ Replace

Cone P/N:

Cup P/N:

Findings:

## Output seals

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

## Housing and bearing bores

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

## Lubrication: breather, plugs, sight glass, oil lines

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

## Bearing caps and shims

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

## Other:

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

## Other:

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

## Other:

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

## G. Failure modes observed

Gears: ☐ Pitting ☐ Spalling ☐ Scuffing/scoring ☐ Wear ☐ Tooth breakage

# Ram Gear Gearbox Evaluation

|                          |                          |                           |                          |                          |                            |                            |
|--------------------------|--------------------------|---------------------------|--------------------------|--------------------------|----------------------------|----------------------------|
| <input type="checkbox"/> | Plastic deformation      | <input type="checkbox"/>  | Corrosion                | <input type="checkbox"/> | Misaligned contact pattern |                            |
| <b>Bearings:</b>         | <input type="checkbox"/> | Spalling/fatigue          | <input type="checkbox"/> | Brinelling               | <input type="checkbox"/>   | Contamination dents        |
|                          | <input type="checkbox"/> | Overheating/discoloration | <input type="checkbox"/> | Corrosion/etching        | <input type="checkbox"/>   | Electrical fluting         |
|                          | <input type="checkbox"/> | Cage damage               | <input type="checkbox"/> | Skidding                 |                            |                            |
| <b>Planetary:</b>        | <input type="checkbox"/> | Planet pin wear           | <input type="checkbox"/> | Needle roller damage     | <input type="checkbox"/>   | Thrust washer wear         |
|                          | <input type="checkbox"/> | Carrier cracks            | <input type="checkbox"/> | Spline fretting          | <input type="checkbox"/>   | Ring gear loose in housing |

Location / notes

## H. Dimensional checks

### Bearing bore and journal diameters

| Location                | Bearing bore measured | Bearing bore spec    | Bore OK                  | Journal measured     | Journal spec         | Journal OK               |
|-------------------------|-----------------------|----------------------|--------------------------|----------------------|----------------------|--------------------------|
| Input shaft bearing     | <input type="text"/>  | <input type="text"/> | <input type="checkbox"/> | <input type="text"/> | <input type="text"/> | <input type="checkbox"/> |
| Output shaft bearing    | <input type="text"/>  | <input type="text"/> | <input type="checkbox"/> | <input type="text"/> | <input type="text"/> | <input type="checkbox"/> |
| Stage 1 carrier bearing | <input type="text"/>  | <input type="text"/> | <input type="checkbox"/> | <input type="text"/> | <input type="text"/> | <input type="checkbox"/> |
| Stage 2 carrier bearing | <input type="text"/>  | <input type="text"/> | <input type="checkbox"/> | <input type="text"/> | <input type="text"/> | <input type="checkbox"/> |
| Stage 3 carrier bearing | <input type="text"/>  | <input type="text"/> | <input type="checkbox"/> | <input type="text"/> | <input type="text"/> | <input type="checkbox"/> |

Seal journal condition notes

## I. Root cause

|                          |                       |                          |                       |                          |                       |                          |              |
|--------------------------|-----------------------|--------------------------|-----------------------|--------------------------|-----------------------|--------------------------|--------------|
| <input type="checkbox"/> | Lubrication failure   | <input type="checkbox"/> | Contamination         | <input type="checkbox"/> | Overload / shock load | <input type="checkbox"/> | Misalignment |
| <input type="checkbox"/> | Improper installation | <input type="checkbox"/> | Fatigue / end of life | <input type="checkbox"/> | Manufacturing defect  |                          |              |

# Ram Gear Gearbox Evaluation

☐ Other

Root cause narrative

## J. Repair summary

Parts to replace

Machining / repair work required

Recommended upgrades or changes

Estimated labor hours

Estimated parts cost

Quote number

**Recommendation:** ☐ Repair ☐ Replace unit ☐ Not economical to repair

## K. Sign-off

**Inspected by**

Name

Signature (type name)

Date

**Reviewed / approved by**

Name

Signature (type name)

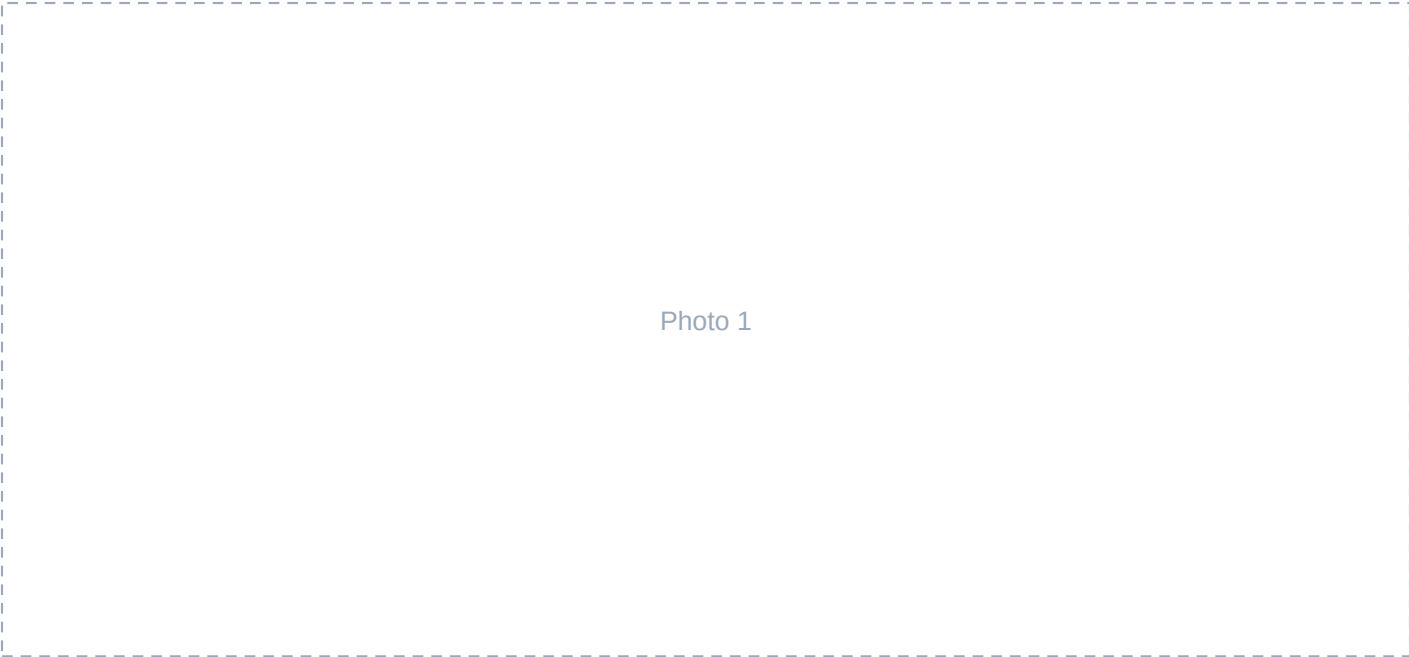
Date

☐ Customer approval received

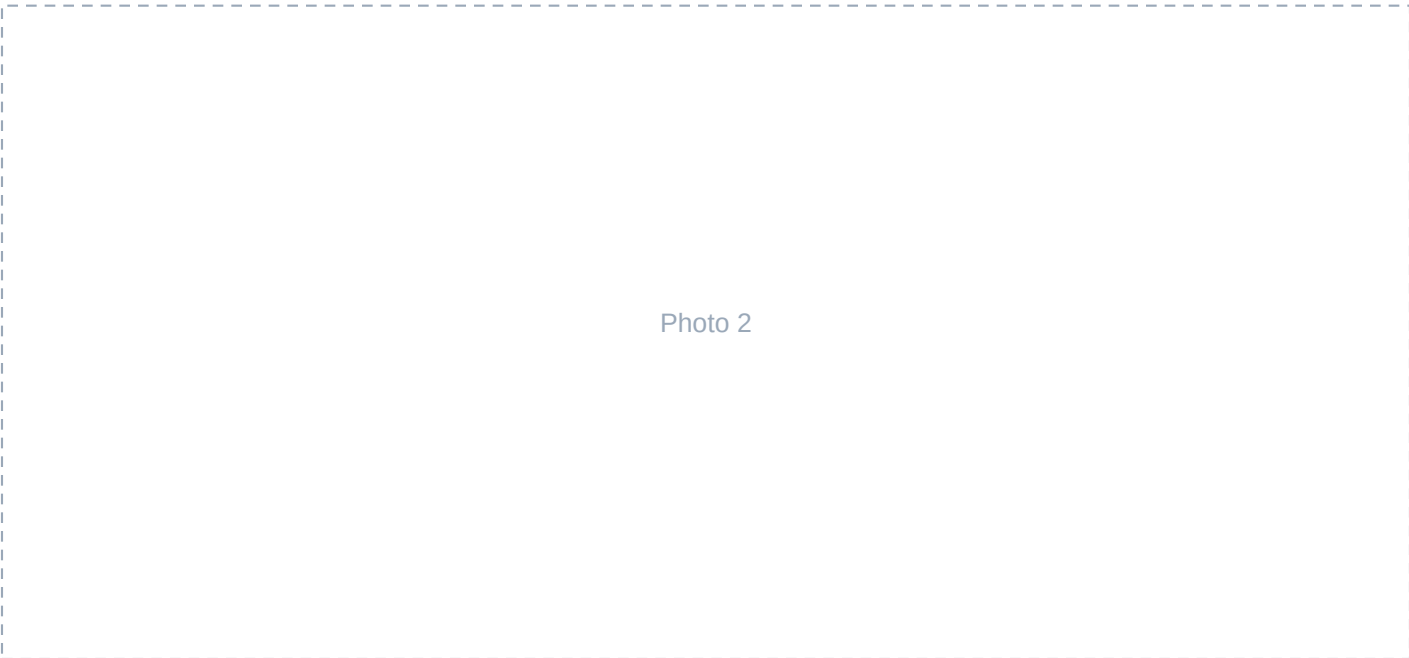
Customer approval date

## L. Teardown photos

Attach teardown photos. Label each photo with a short caption, e.g. As received, Disassembled or a component name. Forms completed in the app print the photos and captions in this section.



Caption: \_\_\_\_\_



Caption: \_\_\_\_\_